Principia Orthogona

Volume One: The Mathematics of Generative Transitions

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Dedicated to my children Vic (R.I.P.), Giulia, Alice (R.I.P.), Sarah (R.I.P.), and David.

Once tiny, always strong.

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Preface

This volume develops a unified mathematical framework for generative transitions: localized geometric events in which a trajectory undergoes compression, curvature intensification, loss of injectivity, and stabilization.

The central object is the operator sequence

$$C \;\to\; K \;\to\; F \;\to\; U,$$

acting on trajectories in a Riemannian manifold.

Several results are presented with proof sketches rather than complete proofs. These include the symplectic preservation across the fold, the classification theorem for admissible singularities, the existence and uniqueness of the unfolding map, and the stability of κ^* under perturbations. Each can be completed with standard tools from differential geometry, singularity theory, and variational analysis; full proofs lie beyond the scope of this foundational volume.

This volume does not claim that all physical, biological, cognitive, or economic systems instantiate this structure. It provides the mathematical substrate on which such questions can be posed rigorously. Cross-domain instantiation is the subject of Volume Three.

This volume is the first in a three-part series. The subsequent papers—*Generative Contact Mechanics: A Geometric Framework for Dissipative Systems with Structured Limit Cycles* and *The dm3 Operator: Explicit Toy Model and Global Dynamical Analysis*—develop the full contact-geometric realization of the generative transition theory introduced here.

1 Overview

Let (X, g) be a smooth Riemannian manifold. A *generative transition* is a localized event along a trajectory $\gamma : [0, T] \to X$ consisting of four sequential phases: compression, curvature induction, folding, and stabilization.

The theory is organized around the composite operator

$$\mathcal{G} = U \circ F \circ K \circ C : X \to X.$$

The manifold $N = (M, \Phi, F, K, g)$ serves as the canonical instance, where M is the geometric substrate, Φ the stability functional, F the directional field, K the constraint operator, and g the generative operator.

The generative flow on M is governed by

$$\frac{dx}{dt} = -\nabla\Phi(x) + F(x) + \eta(t),$$

where $\eta(t)$ is a stochastic perturbation.

2 Minimal Mathematical Assumptions

The operator sequence is defined under the weakest assumptions necessary for the constructions of Section 3 to be well-posed.

Assumption 2.1 (Manifold Structure)**.** *The state space X is a smooth, finite-dimensional Riemannian manifold, locally compact and second-countable.*

Assumption 2.2 (Trajectory Regularity)**.** *A system trajectory $\gamma : [0, T] \to X$ is piecewise C^2, locally non-degenerate ($\|\dot{\gamma}(t)\| \neq 0$ a.e.), and has bounded curvature on compact intervals prior to folding events.*

Assumption 2.3 (Compression Feasibility)**.** *There exists a lower-dimensional submanifold $X_C \subset X$ and a Lipschitz continuous projection $C : X \to X_C$ satisfying the non-collapse condition*

$$d\big(C(x_1), C(x_2)\big) \;\geq\; \delta\, d(x_1, x_2)$$

for some $\delta > 0$ and all x_1, x_2 in a compact neighborhood.

Remark 2.1. Assumption 2.3 is a bi-Lipschitz embedding condition on the compression map, ensuring distinct trajectories remain distinguishable after compression. This is stronger than mere Lipschitz continuity and should be understood as a lower bound on distance preservation.

Assumption 2.4 (Curvature Threshold)**.** *The critical curvature is defined intrinsically by the focal radius:*

$$\kappa^*(x) \;=\; \frac{1}{\mathrm{foc}(x)},$$

where $\mathrm{foc}(x)$ *is the distance from x to the first focal point along the normal direction. In the presence of positive sectional curvature $K_{\mathrm{sec}}(x) > 0$, this is modified by the Rauch comparison theorem to*

$$\kappa^*(x) \;=\; \min\Big(\|II_x\|,\; \sqrt{K_{\mathrm{sec}}(x)}\Big),$$

where $\|II_x\|$ is the norm of the second fundamental form. For $K_{\mathrm{sec}}(x) \leq 0$, we have $\kappa^*(x) = \|II_x\|$.

Assumption 2.5 (Folding Well-Posedness)**.** *The folding operator $F : X_C \to X_F$ satisfies: the Jacobian dF loses rank by exactly* 1 *at fold points; the fold is local; and the fold produces a finite number of branches.*

Assumption 2.6 (Morse Stability Functional)**.** *The stability functional $\Phi : X \to \mathbb{R}$ is C^2, bounded below on compact subsets, and* Morse*: all critical points are non-degenerate, i.e., $\nabla^2\Phi(x^*) \succ 0$ at every local minimum x^*.*

3 Operator Definitions

3.1 State Space

Let X be as in Assumption 2.1. A system trajectory is a piecewise C^2 map $\gamma : [0, T] \to X$. The goal is to describe local transitions in γ via the sequence $C \to K \to F \to U$.

3.2 Compression Operator C

Definition 3.1 (Compression)**.** *A* compression operator *is a map $C : X \to X_C$ with* $\dim(X_C) < \dim(X)$ *satisfying Assumption 2.3.*

C reduces degrees of freedom while retaining essential local structure.

3.3 Curvature Operator K

Definition 3.2 (Curvature Operator)**.** *Given a compressed trajectory $\gamma_C : [0, T] \to X_C$, the curvature operator $K : X_C \to X_C$ modifies the tangent field by*

$$\frac{d}{ds}\big(K(\gamma_C)(s)\big) \;=\; \dot{\gamma}_C(s) + \alpha(s)\, n(s),$$

where $n(s)$ is the unit normal and

$$\alpha(s) \;=\; \lambda\left(\kappa^*(\gamma_C(s)) - \kappa(s)\right)_+, \quad \lambda > 0.$$

K is a curvature-inducing flow that drives curvature monotonically toward κ^* but never beyond it. Consequently, K alone does not trigger folding; the sequence is a *hybrid dynamical system* with a switching condition at κ^*.

3.4 Folding Operator F

Definition 3.3 (Folding Map)**.** *Let $\gamma_K = K(\gamma_C)$. A* fold *occurs at s_0 when $|\kappa_K(s_0)| = \kappa^*(\gamma_K(s_0))$. The folding operator $F : X_C \to X_F$ acts by*

$$F(\gamma_K(s)) \;=\; \gamma_K(s) - \beta(s)\, n(s),$$

where

$$\beta(s) \;=\; \begin{cases} 0, & |\kappa_K(s)| < \kappa^*, \\ \mu\big(|\kappa_K(s)| - \kappa^*\big), & |\kappa_K(s)| \geq \kappa^*. \end{cases}$$

Here $\mu > 0$ controls fold depth.

At a fold point s_0, $\operatorname{rank}(dF_{\gamma_K(s_0)}) = \dim(X_C) - 1$.

3.5 Unfolding Operator U

Definition 3.4 (Unfolding Map). *The unfolding operator $U : X_F \to X$ is*

$$U(x_F) \;=\; \arg\min_{y \in \mathcal{N}(x_F)} \Phi(y),$$

where $\mathcal{N}(x_F)$ is a sufficiently small neighborhood of x_F and Φ satisfies Assumption 2.6.

The unfolding process is realised by the gradient flow $\dot{y} = -\nabla\Phi(y)$, $y(0) = x_F$, which converges to a non-degenerate local minimum x^*.

3.6 Compatibility of K and F

Theorem 3.1 (Sequential Consistency). *If K is applied until curvature reaches κ^*, then F is well-defined, produces a finite branch set, and induces a rank-deficient Jacobian at fold points.*

Proof sketch. K drives curvature monotonically to κ^* via the gain field $\alpha(s)$. At κ^*, the term $\beta(s)$ becomes nonzero, introducing local non-injectivity. The normal direction collapses, reducing Jacobian rank by 1. Locality and the finite critical-point structure of κ^* (via the Morse assumption on Φ) imply finitely many branches. □

4 Falsifiability Conditions

The framework is falsifiable at four levels.

Compression. If empirical data show expansion under C, or collapse of distinct trajectories, the model fails.

Curvature. If a fold occurs at curvature strictly below κ^*, or curvature exceeds κ^* without folding, the model is falsified. (κ^* is computed independently via the focal radius, not post-hoc from fold observations.)

Folding. If empirical fold events do not correspond to Jacobian rank loss, or produce infinitely many branches, the model fails.

Sequence order. If transitions occur in a different order, or stabilization occurs without folding, the model is falsified.

5 Structural Theorems

Theorem 5.1 (Existence and Well-Posedness)**.** *Under Assumptions 2.1–2.6, the composite operator* $\mathcal{G} = U \circ F \circ K \circ C$ *is well-defined on any piecewise* C^2 *trajectory.*

Theorem 5.2 (Local Determination)**.** *The action of* $\mathcal{G}$ *on* γ *is determined entirely by the local geometry of* γ *in a neighborhood of the fold point.*

Theorem 5.3 (Non-Commutativity)**.** *The operators* C, K, F, U *do not commute; the sequence is order-dependent.*

Theorem 5.4 (Irreducibility)**.** *No operator in the sequence* $C \to K \to F \to U$ *can be removed without altering the qualitative structure of the transition.*

Theorem 5.5 (Finite Branching)**.** *The branch set* $B = \{F(\gamma_K(s_i)) : |\kappa_K(s_i)| = \kappa^*(\gamma_K(s_i))\}$ *is finite.*

Remark 5.1. Finiteness follows from the Morse condition on Φ (Assumption 2.6), which prevents fourth-order degeneracies and ensures isolated critical points, together with the transversality of γ to the fold locus (a generic condition to be assumed explicitly in applications).

6 Canonical Examples

Planar curve with curvature-driven flow. The simplest nontrivial system: $\gamma : [0, T] \to \mathbb{R}^2$, with C an orthogonal projection, K the curvature-inducing flow, F activated at κ^*, and U gradient descent on a local Φ.

Elastic rod under compression. An elastic rod minimising Euler–Bernoulli energy $E[\gamma] = \int \kappa(s)^2 \, ds$. Axial loading provides C; buckling provides K; the critical load is κ^*; post-buckling configuration provides U. This is the cleanest physical realisation of the curvature threshold.

Gradient flow on a double-well potential. $\Phi(x) = (x^2 - 1)^2$. The unstable equilibrium at $x = 0$ is the fold point; gradient flow selects one of $x = \pm 1$. Illustrates finite branching and stability selection.

Saddle-node bifurcation. $\dot{x} = \mu - x^2$, $\dot{y} = -y$. Projection onto the slow manifold provides C; approach to the fold point provides K; loss of the slow manifold at $\mu = 0$ provides F; flow to the stable branch provides U.

Plasma sheet and market manifolds (Volume Three). Magnetic field-line reconnection and price-trajectory bifurcations are conjectured instantiations. Each requires domain-specific construction of (X, g, Φ) and empirical computation of κ^*. These are deferred to Volume Three.

7 Analytical Invariants

Invariant 7.1 (Ambient Dimension). $\dim(X)$ *is preserved by* $\mathcal{G}$. *No claim is made about the dimension of the image of a specific trajectory.*

Invariant 7.2 (Codimension of the Fold). $\operatorname{codim}(F(\gamma)) = 1$, *following from the rank-loss condition.*

Invariant 7.3 (Critical Curvature Threshold). $\kappa^*(x)$ *is a geometric property of the manifold, invariant under the operator sequence.*

Invariant 7.4 (Curvature Sign). $\operatorname{sgn}(\kappa(s))$ *is preserved under* K *and* F*; only magnitude changes.*

Invariant 7.5 (Injectivity Before Threshold). *For all* s *with* $|\kappa(s)| < \kappa^*(s)$, *the trajectory remains injective under* C, K, F.

Invariant 7.6 (Rank Deficiency at Fold). $\operatorname{rank}(dF) = \dim(X_C) - 1$ *at every fold point.*

Invariant 7.7 (Energy Monotonicity under U). $\Phi(U(x)) \leq \Phi(x)$, *with strict inequality unless* x *is already a local minimum.*

Remark 7.1. An energy-gap invariant $\Delta\Phi = \Phi(x_2) - \Phi(x_1)$ is preserved under C, K, F only if Φ is constant along the fibers of each operator. This is not guaranteed in general and is not claimed as a universal invariant.

8 Normal Forms

Two transitions $\mathcal{G}_1$, $\mathcal{G}_2$ are equivalent if there exists a diffeomorphism $\psi : X \to X$ with $\mathcal{G}_2 = \psi \circ \mathcal{G}_1 \circ \psi^{-1}$.

Normal Form 8.1 (Compression). $C_{\mathrm{NF}} = \pi_k : \mathbb{R}^n \to \mathbb{R}^k$, *the standard coordinate projection.*

Normal Form 8.2 (Curvature). $K_{\mathrm{NF}}(s) = (s, \alpha s^2)$, $\alpha > 0$. *This produces curvature* $\kappa(s) = 2\alpha(1 + 4\alpha^2 s^2)^{-3/2}$, *which approximates* 2α *near* $s = 0$.

Normal Form 8.3 (Folding — Whitney Fold). $F_{\mathrm{NF}}(u, v) = (u, v^2)$. *This is the unique (up to diffeomorphism) normal form for a codimension-1 fold singularity.*

Normal Form 8.4 (Unfolding). $U_{\mathrm{NF}}(x) = 0$, *the canonical stabilization map arising from* $\Phi_{\mathrm{NF}}(x) = x^2$.

9 Singularity Classification

9.1 Admissible Singularities

Given the constraints of rank-1 loss, finite branching, and the Morse condition on Φ, the admissible singularities are precisely:

1. Fold A_1: $(u, v) \mapsto (u, v^2)$
2. Cusp A_2: $(u, v) \mapsto (u, v^3 + uv)$
3. Swallowtail A_3: $(u, v) \mapsto (u, v^4 + uv^2 + \beta v)$

Higher A_k ($k \geq 4$) require a k-parameter unfolding. The Morse condition on Φ restricts the unfolding to at most three parameters, excluding A_4 and above.

9.2 Classification Conditions

Let $\Delta(s) = \kappa(s) - \kappa^*(s)$.

Type	Conditions	Normal form
A_1 (fold)	$\Delta(s_0) = 0$, $\Delta'(s_0) \neq 0$	(u, v^2)
A_2 (cusp)	$\Delta = \Delta' = 0$, $\Delta'' \neq 0$	$(u, v^3 + uv)$
A_3 (swallowtail)	$\Delta = \Delta' = \Delta'' = 0$, $\Delta''' \neq 0$	$(u, v^4 + uv^2 + \beta v)$

Theorem 9.1 (Classification of Generative Transitions)**.** *Every admissible generative transition $\mathcal{G} = U \circ F \circ K \circ C$ is $\mathcal{A}$-equivalent to exactly one of A_1, A_2, A_3.*

Proof sketch. Rank loss is exactly 1, restricting to the A_k series. The Morse condition on Φ limits the unfolding to at most three parameters. Transversality of γ to the fold locus (generic assumption) ensures isolated fold points. Thus $k \leq 3$. □

9.3 Bifurcation Stratification

Parameter space $\Theta \subset \mathbb{R}^p$, $p \leq 3$, decomposes into:

- Θ_1: generic fold region (codimension 0 in Θ),
- Θ_2: cusp stratum (codimension 1),
- Θ_3: swallowtail stratum (codimension 2; a single point when $p = 3$).

10 Metric Geometry of κ^*

10.1 Intrinsic Definition

$$\kappa^*(x) \;=\; \frac{1}{\mathrm{foc}(x)},$$

where $\mathrm{foc}(x)$ is the focal radius. For a curve embedded in (X, g), this equals $\|II_x\|$ (the norm of the second fundamental form) when $K_{\mathrm{sec}} \leq 0$.

10.2 Sectional Curvature Correction

By the Rauch comparison theorem, for variable positive sectional curvature:

$$\kappa^*(x) \;=\; \min\Big(\|II_x\|,\; \sqrt{K_{\mathrm{sec}}(x)}\Big).$$

Positive ambient curvature lowers the threshold for folding.

10.3 Computability

Given γ in (X, g):

1. Compute $\|II_x\|$ from the second fundamental form.
2. Compute $K_{\mathrm{sec}}(x)$ from the curvature tensor.
3. Apply the formula above.

κ^* is stable: $\delta\kappa^* = O(\delta g) + O(\delta II) + O(\delta K_{\mathrm{sec}})$.

11 Variational Principles

Define the action functional

$$\mathcal{S}[\gamma] \;=\; \int_0^T \Big(\tfrac{1}{2}\|P_\perp \dot{\gamma}\|^2 + \tfrac{\lambda}{2}(\kappa^* - \kappa)_+^2 + \mu\,\delta(|\kappa| - \kappa^*) + \Phi(\gamma)\Big)\, ds.$$

Each term corresponds to one operator:

- $L_C = \frac{1}{2}\|P_\perp \dot{\gamma}\|^2$: minimise orthogonal kinetic energy (compression).
- $L_K = \frac{\lambda}{2}(\kappa^* - \kappa)_+^2$: minimise curvature deficit.
- $L_F = \mu\,\delta(|\kappa| - \kappa^*)$: singular activation at threshold.
- $L_U = \Phi(\gamma)$: minimise stability potential.

The delta term places the framework in the class of *free-discontinuity variational problems* (cf. Ambrosio–Tortorelli, Mumford–Shah, Francfort–Marigo). It produces the jump condition

$$\left[\frac{\partial L}{\partial \dot{\gamma}}\right]_{s_0^-}^{s_0^+} = \mu\, n(s_0),$$

which is the variational counterpart of the fold map.

The full generative transition is therefore a *piecewise-smooth extremal* of $\mathcal{S}$.

12 Hamiltonian Discontinuities and Symplectic Geometry

12.1 Phase Space

The phase space is T^*X with canonical coordinates (γ, p) and symplectic form $\omega = d\gamma \wedge dp$.

12.2 Smooth Flow

For $s \neq s_0$, the system obeys Hamilton's equations and the flow is symplectic: $\mathcal{F}_t^*\omega = \omega$.

12.3 Momentum Jump at the Fold

The delta Lagrangian produces the impulse

$$p(s_0^+) - p(s_0^-) \;=\; \mu\, n(s_0).$$

Configuration γ is continuous; momentum p has a jump.

12.4 The Fold as a Symplectic Map

Define the fold map in phase space: $\mathcal{F} : (\gamma, p) \mapsto (\gamma, p + \mu n)$.

Theorem 12.1 (Symplectic Preservation)**.** *$\mathcal{F}^*\omega = \omega$.*

Proof. $d\gamma \wedge d(p + \mu n) = d\gamma \wedge dp + \mu\, d\gamma \wedge dn$. Since n depends only on γ, the term $d\gamma \wedge dn = 0$. Hence $\mathcal{F}^*\omega = \omega$. □

12.5 Canonical Transformation

The fold is generated by the distributional function $S(\gamma) = \mu\, \Theta(|\kappa(\gamma)| - \kappa^*)$, so that $p^+ = p^- + \partial S/\partial\gamma$.

12.6 Full Transition

Let Φ_t be the pre-fold Hamiltonian flow, $\mathcal{F}$ the fold map, Ψ_t the post-fold flow. The consistency condition from Section 4 (basin structure) ensures the output of $\mathcal{F}$ lies in the domain of Ψ_t. Then

$$\mathcal{H} = \Psi_t \circ \mathcal{F} \circ \Phi_t$$

is a piecewise-smooth symplectic map: $\mathcal{H}^*\omega = \omega$.

12.7 Singularity Type and Momentum Structure

- A_1 fold: single momentum jump.
- A_2 cusp: momentum jump with first-order tangency constraint.
- A_3 swallowtail: momentum jump with second-order tangency.

Closing Statement

This volume has developed a unified mathematical framework for generative transitions. The central results are:

- a geometric foundation based on curvature, injectivity radius, and focal geometry;
- a critical curvature threshold κ^* defined intrinsically by the focal radius and corrected by the Rauch comparison theorem;
- constructive definitions of operators C, K, F, U;
- a free-discontinuity variational principle;
- a Hamiltonian structure with a distributional generator at the fold;
- a symplectic discontinuity preserving the canonical form;
- a singularity classification restricted to the Whitney A_1–A_3 hierarchy; and
- structural invariants holding across all admissible transitions.

The framework is placed within established mathematical traditions: comparison geometry, singularity theory, geometric flows, variational mechanics, and impulsive Hamiltonian systems.

This volume is deliberately limited in scope. It provides the mathematical substrate on which cross-domain questions can be posed rigorously. Volume Two develops discrete variational integrators and numerical realisation. Volume Three instantiates the framework in specific domains: plasma reconnection, market volatility manifolds, and neural embedding geometry.

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This volume is the first in a series. The dm3 operator framework developed in the companion papers *Generative Contact Mechanics* (Paper 1) and *The dm3 Operator: Explicit Toy Model* (Paper 2) constitutes the full mathematical realization of the generative transition theory introduced here.